

DEPARTMENT OF CHEMISTRY, INSTITUTE OF SCIENCE, BHU  
B.Sc. (Hons.) I Semester (NEP-2020) Mid Semester Test: 2024-2025

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Basic Concepts of Chemistry-I (Physical Chemistry)

Date: 20 Dec 2024

Maximum Marks: 13

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*Note: Attempt all the questions. Marks for each question are given in the margin.*

**Question 1.** Complete the following: [3]

- (a) The expression for the law of corresponding states.....
  - (b) The expression of the Poiseuille equation.....
  - (c) Eotvos's empirical relation between surface tension and temperature.....
  - (d) The expression of collision frequency in terms of pressure....
  - (e) For the orthorhombic crystal system, axial ratio..... and angles.....
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**Question 2.** Derive Graham's law of diffusion using the kinetic gas equation. [1]

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**Question 3.** What is the excluded volume? Show that the excluded volume is four times the actual volume of a molecule. [2]

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**Question 4.** How will you determine critical volume experimentally? [2]

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**Question 5.** Discuss the effect of temperature on viscosity with the help of hole theory. [2]

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**Question 6.** Explain the steps in converting Weiss indices into Miller indices with a suitable example. [2]

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**Question 7.** Draw and explain the structure of the diamond. [1]

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